

Input Pair



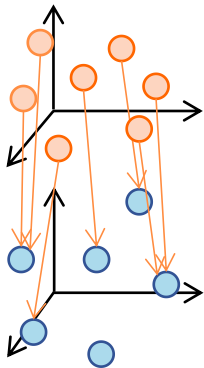
Reference
Point Cloud



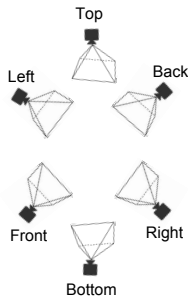
Distorted
Point Cloud

Pre-processing

Point Alignment



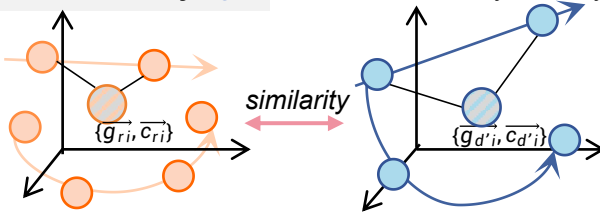
2D Projection



Multimodal Perceptual Representation

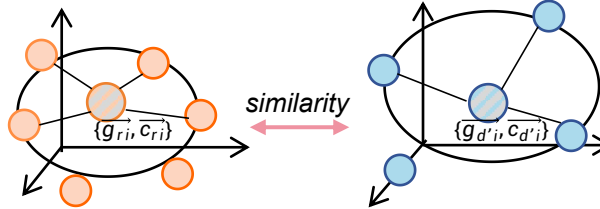
3D Geometry F_G

curvature, linearity, density



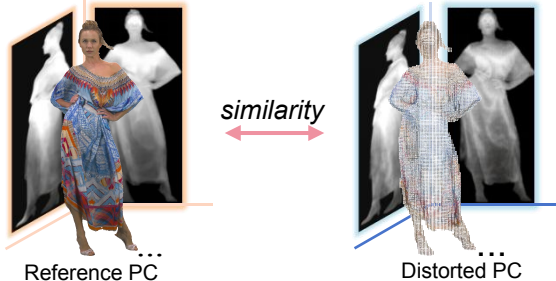
3D Color F_C

color contrast



2D Saliency F_S

saliency map



Perceptual Cue Fusion

Stage I: Intrinsic 3D Fusion

$$F_G, F_C \rightarrow F_{3D}$$

Stage II: 2D-3D Fusion

$$F_S, F_{3D} \rightarrow F$$

Final Quality Score

